

Module 2
Vectors, Vector Spaces & Linear Algebra
Complete Study Notes

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Module Outline

- ① Algebraic Structures
- ② Vector Spaces
- ③ Coordinate Systems & Vectors
- ④ Vector Addition & Scalar Multiplication
- ⑤ Linear Combinations, Span, Basis
- ⑥ Matrix Multiplication as Composition
- ⑦ Three-Dimensional Linear Transformations
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- ⑨ Inverse Matrices, Column Space, Null Space
- ⑩ Nonsquare Matrices as Transformations
- ⑪ Dot Products and Duality
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- ⑬ Cross Products & Linear Transformations
- ⑭ Cramer's Rule
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- ⑰ Hyperplanes & Weight Vectors
- ⑱ Metric & Inner Product Spaces
- ⑲ Perceptron (Application)

Definition

Let S be a **set** and $*$ a **binary operation** on S . The pair $(S, *)$ is called an **algebraic structure**.

- A binary operation takes two elements of S and combines them to give another element.
- The properties satisfied by $*$ determine what kind of algebraic structure $(S, *)$ is.
- This gives a hierarchy: **Magma** $\rightarrow$ **Semigroup** $\rightarrow$ **Monoid** $\rightarrow$ **Group** $\rightarrow$ **Abelian Group**.

Closure Property

For all $a, b \in S$: $a * b \in S$.

Magma

If $(S, *)$ satisfies the closure property, it is called a **Magma**.

- Closure simply means: combining two elements of the set never takes you *outside* the set.
- Example: $(\mathbb{Z}, +)$ is a magma since the sum of two integers is always an integer.

Associative Property

$$a * (b * c) = (a * b) * c \quad \forall a, b, c \in S$$

Semigroup

A magma $(S, *)$ that also satisfies associativity is called a **Semigroup**.

- The order in which we group operations does not matter.
- Example: $(\mathbb{N}, +)$ is a semigroup.

Identity Element

There exists $e \in S$ such that

$$a * e = e * a = a \quad \forall a \in S$$

Monoid

A semigroup with an identity element is called a **Monoid**.

- e is called the **identity element** of the structure.
- Example: $(\mathbb{Z}, +)$ is a monoid with identity $e = 0$.
- Example: $(\mathbb{Z}, \times)$ is a monoid with identity $e = 1$.

Inverse Element

For every $a \in S$, there exists $a^{-1} \in S$ such that

$$a * a^{-1} = a^{-1} * a = e$$

Group

A monoid in which every element has an inverse is called a **Group**.

- Every element can be "undone" by combining it with its inverse.
- Example: $(\mathbb{Z}, +)$ is a group; inverse of a is $-a$.
- Example: $(\mathbb{Z}, \times)$ is *not* a group (no inverse for most integers).

Commutative Property

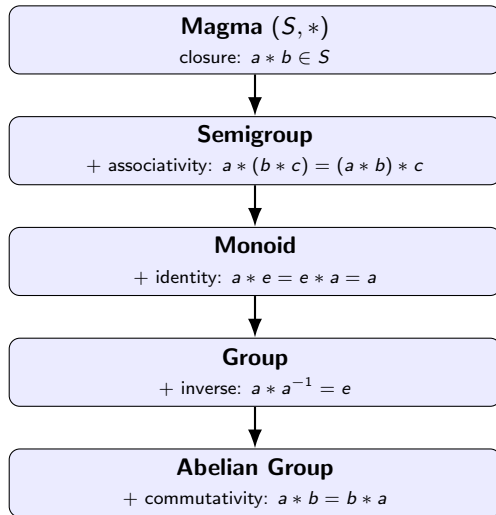
$$a * b = b * a \quad \forall a, b \in S$$

Abelian Group

A group that also satisfies commutativity is called an **Abelian Group**.

- Order of combining two elements does not affect the result.
- Example: $(\mathbb{Z}, +)$ is an abelian group.
- Example: The set of $n \times n$ invertible matrices under multiplication is a group but *not* abelian (in general).

Hierarchy of Algebraic Structures



Ring

A set with **two** binary operations (usually $+$ and $\times$) such that:

- $(S, +)$ is an abelian group,
- $(S, \times)$ is a semigroup (associative),
- multiplication distributes over addition.

Field

A ring in which $(S \setminus \{0\}, \times)$ is also an abelian group (every nonzero element has a multiplicative inverse).

- Example of a field: $(\mathbb{R}, +, \times)$, $(\mathbb{Q}, +, \times)$, $(\mathbb{C}, +, \times)$.
- A **Vector Space** is built from a **Field** F (scalars) and a set V (vectors) together with vector addition and scalar multiplication: $(V, F, +, \cdot)$.

Quick Recap Table: Algebraic Structures

Structure	Closure	Assoc.	Identity	Inverse
Magma	✓			
Semigroup	✓	✓		
Monoid	✓	✓	✓	
Group	✓	✓	✓	✓
Abelian Group	✓	✓	✓	✓*

* Abelian Group additionally requires **commutativity**: $a * b = b * a$.

What is a Vector Space?

Structure $(V, F, +, \cdot)$

- V = set of **vectors**
- F = **scalar field** (e.g. $\mathbb{R}$)
- $+$: **vector addition** ($V \times V \rightarrow V$)
- $\cdot$: **scalar multiplication** ($F \times V \rightarrow V$)

Definition

The **set of all elements** in a vector space is called the set of **vectors**.

- Any object that can be described using multiple values can be treated as a vector.
- Every polynomial is a vector (in the vector space of polynomials).

Examples of Vector Spaces

- $\mathbb{R}^n$ with usual vector addition and scalar multiplication — the standard example.
- The set of all $m \times n$ matrices under matrix addition and scalar multiplication.
- The set of all polynomials of degree $\leq n$: e.g. $p(x) = a_0 + a_1x + \cdots + a_nx^n$.
- The set of continuous real-valued functions on $[a, b]$, under pointwise addition and scaling.

Key Insight

Anything satisfying the 10 vector space axioms — numbers, matrices, polynomials, or functions — can be treated as a "vector," which is what makes linear algebra so widely applicable.

Vector Space Axioms

For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalars $k, l \in F$:

① $\mathbf{u} + \mathbf{v} \in V$ (closure)

② $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

③ $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$

④ $\exists \mathbf{0} : \mathbf{u} + \mathbf{0} = \mathbf{u}$

⑤ $\exists -\mathbf{u} : \mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

⑥ $k\mathbf{u} \in V$ (closure under scalar mult.)

⑦ $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

⑧ $(k + l)\mathbf{u} = k\mathbf{u} + l\mathbf{u}$

⑨ $k(l\mathbf{u}) = (kl)\mathbf{u}$

⑩ $1 \cdot \mathbf{u} = \mathbf{u}$

- Any object that can be "vectorised" with n -valued vectors and satisfies these axioms forms a vector space.

Key Idea

Any object described using **multiple values** — a **Vector** — will always have a **direction** and a **magnitude**.

Example

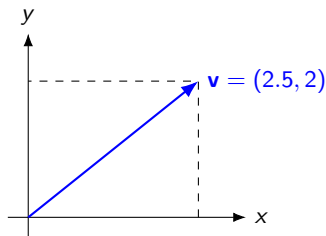
$$V = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad k = 3$$

$$k \cdot V = 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

- **Scalar multiplication changes the magnitude of a vector** (and, if $k < 0$, reverses its direction).

Coordinate System and Vectors

- A **coordinate system** lets us represent a vector as an ordered tuple of numbers (its components) relative to chosen basis directions.
- In $\mathbb{R}^2$: $\mathbf{v} = (x, y)$; in $\mathbb{R}^n$: $\mathbf{v} = (x_1, x_2, \dots, x_n)$.
- Each vector can be visualised as an arrow from the origin to the point given by its coordinates.



Vector Addition (Parallelogram / Triangle Law)

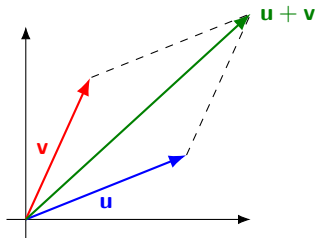
Rule

The **resultant vector** of two vectors becomes the **diagonal of the parallelogram** formed by the two vectors.

Example

$$\mathbf{r}_1 = (3, 5, 2), \quad \mathbf{r}_2 = (1, 3, 4)$$

$$\mathbf{r}_1 + \mathbf{r}_2 = (4, 8, 6)$$



Definition

$$\mathbf{PQ} = \mathbf{Q} - \mathbf{P} = \mathbf{v} - \mathbf{u}$$

If $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$: $\mathbf{PQ} = (v_1 - u_1, v_2 - u_2)$

Squared Norm of $\mathbf{PQ}$

$$\begin{aligned}\|\mathbf{PQ}\|^2 &= (v_1 - u_1)^2 + (v_2 - u_2)^2 \\ &= (v_1^2 + v_2^2) + (u_1^2 + u_2^2) - 2(v_1 u_1 + v_2 u_2) \\ &= \|\mathbf{v}\|^2 + \|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle\end{aligned}$$

This is exactly the **Law of Cosines**: $v^2 + u^2 - 2uv \cos \theta$.

Definition

A **linear combination** of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is any vector of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$$

where $c_1, \dots, c_n$ are scalars.

- Scaling and adding vectors together produces new vectors.
- Every vector in $\mathbb{R}^2$ can be written as a linear combination of $\hat{i} = (1, 0)$ and $\hat{j} = (0, 1)$.

Definition

The **span** of a set of vectors is the set of **all possible linear combinations** of those vectors.

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_n) = \{c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n : c_i \in F\}$$

- Span of one nonzero vector in $\mathbb{R}^2 \rightarrow$ a line through the origin.
- Span of two non-parallel vectors in $\mathbb{R}^2 \rightarrow$ the entire plane $\mathbb{R}^2$.
- If a vector's direction can be produced by scaling and adding others, it lies in their span.

Definition

A **basis** of a vector space V is a set of **linearly independent** vectors that **span** V .

- **Linear independence**: no vector in the set can be written as a linear combination of the others.
- Every vector in V can be written **uniquely** as a linear combination of basis vectors.
- **Standard basis** of $\mathbb{R}^n$: $\{e_1, e_2, \dots, e_n\}$ where e_i has 1 in position i and 0 elsewhere.
- The number of vectors in a basis = **dimension** of the vector space.

Worked Example: Linear Combination

Problem

Express $\mathbf{v} = (7, 4)$ as a linear combination of $\mathbf{v}_1 = (2, 1)$ and $\mathbf{v}_2 = (1, 2)$.

Solution

Solve $c_1(2, 1) + c_2(1, 2) = (7, 4)$:

$$2c_1 + c_2 = 7, \quad c_1 + 2c_2 = 4$$

Solving: $c_1 = 10/3$, $c_2 = 1/3$ (approx.), or use exact elimination to get the required scalars.

- Since $\mathbf{v}_1, \mathbf{v}_2$ are linearly independent and span $\mathbb{R}^2$, every vector in $\mathbb{R}^2$ can be written this way — confirming $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis.

Example: Span and Basis in $\mathbb{R}^2$

Standard Basis

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Any $\mathbf{v} = (x, y) = x \mathbf{e}_1 + y \mathbf{e}_2$.

Another Basis

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

These are linearly independent and span $\mathbb{R}^2$, so they also form a valid basis.

- A matrix A represents a **linear transformation** $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, where $T(\mathbf{x}) = A\mathbf{x}$.
- Columns of A tell us where the basis vectors land after the transformation.

Matrix Multiplication = Composition of Transformations

If A represents transformation T_1 and B represents T_2 , then the product AB represents applying T_2 **first**, then T_1 :

$$(AB)\mathbf{x} = A(B\mathbf{x}) = T_1(T_2(\mathbf{x}))$$

- Order matters: $AB \neq BA$ in general, since composing transformations in a different order gives a different result.

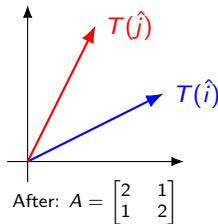
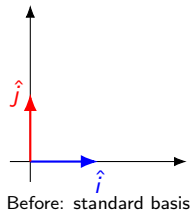
Rotation then Shear

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ (shear),} \quad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ (90}^\circ \text{ rotation)}$$

$$AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

- Applying AB to x is the same as first rotating x by B , then shearing the result by A .

Visualising a Linear Transformation



- The columns of the transformation matrix show exactly where each basis vector lands.
- Every other vector transforms accordingly, since T is linear: $T(\mathbf{x}) = A\mathbf{x}$.

Three-Dimensional Linear Transformations

- A linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is represented by a 3×3 matrix.
- Each column shows where $\hat{i}, \hat{j}, \hat{k}$ (standard basis vectors) land.

General Form

$$T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

where columns (a, d, g) , (b, e, h) , (c, f, i) are the images of $\hat{i}, \hat{j}, \hat{k}$.

- Composition of 3D transformations again corresponds to matrix multiplication.
- The determinant of the 3×3 matrix gives the **volume scaling factor** of the transformation.

The Determinant

Meaning

The **determinant** of a matrix measures how much a linear transformation **scales area (2D)** or **volume (3D)**.

- $\det(A) = 0 \iff$ the transformation squashes space into a lower dimension (columns are linearly dependent).
- Negative determinant $\Rightarrow$ orientation of space is flipped.

2×2 Determinant

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(A) = ad - bc$$

3×3 Determinant (cofactor expansion)

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

- $\det(AB) = \det(A) \det(B)$
- $\det(A^{-1}) = \frac{1}{\det(A)}$ (only exists if $\det(A) \neq 0$)
- $\det(A^T) = \det(A)$
- Swapping two rows/columns flips the sign of the determinant.
- A matrix is **invertible** $\iff \det(A) \neq 0$.

Definition

A^{-1} is the matrix such that

$$AA^{-1} = A^{-1}A = I$$

- Applying A^{-1} **undoes** the transformation done by A .
- A^{-1} exists only when $\det(A) \neq 0$ (i.e. A is **nonsingular**).
- Used to solve linear systems: $A\mathbf{x} = \mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b}$.

2×2 Inverse Formula

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Definition

The **column space** of A , $\text{Col}(A)$, is the **span of the columns** of A — i.e., the set of all possible outputs Ax .

- Tells us the set of vectors "reachable" by the transformation.
- $\dim(\text{Col}(A)) = \text{rank}(A)$.
- If $\text{rank}(A) = n$ (full column rank), the transformation is **injective** (one-to-one).

Definition

The **null space** of A , $\text{Null}(A)$, is the set of all vectors that map to the **zero vector**:

$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$$

- Represents the vectors that get "squashed" to the origin.
- If $\text{Null}(A) = \{\mathbf{0}\}$ only, the transformation is injective.

Rank–Nullity Theorem

$$\text{rank}(A) + \text{nullity}(A) = n \quad (\text{number of columns of } A)$$

Worked Example: Column Space & Null Space

Problem

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Solution

- Columns $(1, 2)$ and $(2, 4)$ are parallel $\Rightarrow \text{rank}(A) = 1$.
- $\text{Col}(A) = \text{span}\{(1, 2)\}$ — a line through the origin.
- $\text{Null}(A)$: solve $Ax = 0 \Rightarrow x_1 = -2x_2 \Rightarrow \text{Null}(A) = \text{span}\{(-2, 1)\}$.
- Check rank-nullity: $\text{rank}(A) + \text{nullity}(A) = 1 + 1 = 2 = n$. ✓

Nonsquare Matrices: Transformations Between Dimensions

- An $m \times n$ matrix represents a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$.
- If $m > n$: maps a **lower-dimensional** space into a **higher-dimensional** one (e.g. $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ embeds a plane in 3D space).
- If $m < n$: maps a **higher-dimensional** space into a **lower-dimensional** one (e.g. $\mathbb{R}^3 \rightarrow \mathbb{R}^2$ projects 3D onto a plane).

Example: $\mathbb{R}^2 \rightarrow \mathbb{R}^3$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \\ x + y \end{bmatrix}$$

The determinant is not defined for nonsquare matrices; instead we talk about **rank**.

Definition

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

- Measures how much two vectors point in the **same direction**.
- $\mathbf{u} \cdot \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v}$ (**orthogonal**).
- $\mathbf{u} \cdot \mathbf{v} > 0$: vectors point in similar direction; < 0 : opposite direction.

Perpendicular Line/Plane Rule

For line $ax + by + c = 0$: normal vector is (a, b) .

For plane $ax + by + cz + d = 0$: normal vector is (a, b, c) .

Scalar Projection of $\mathbf{u}$ onto $\mathbf{v}$

$$\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}$$

Vector Projection

$$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$$

- Gives the "shadow" of $\mathbf{u}$ falling along the direction of $\mathbf{v}$.
- Used to decompose a vector into components parallel and perpendicular to another vector.

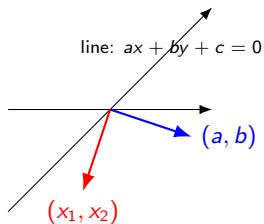
Duality Principle

Every linear transformation from $\mathbb{R}^n \rightarrow \mathbb{R}$ (a $1 \times n$ matrix) corresponds to a unique vector $\mathbf{w}$ such that the transformation is the dot product with $\mathbf{w}$:

$$T(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} = \mathbf{w}^T \mathbf{x}$$

- This is why $1 \times n$ matrices (row vectors) and n -dimensional vectors are "dual" to each other.
- Every dot product can be viewed as first applying the linear transformation $\mathbf{w}^T$, then reading off the single number result.

Perpendicularity via Dot Product



Key Relations

- Line: $ax + by + c = 0 \Rightarrow \mathbf{w} = (a, b) \perp$ the line's direction vector.
- Hyperplane in $\mathbb{R}^n$: $\sum_{i=1}^n w_i x_i + \theta = 0 \Leftrightarrow \mathbf{w}^T \mathbf{x} + \theta = 0$

Definition (in $\mathbb{R}^3$)

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

- Produces a vector **perpendicular to both $\mathbf{u}$ and $\mathbf{v}$** (right-hand rule gives direction).
- Magnitude: $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin \theta =$ **area of the parallelogram** spanned by $\mathbf{u}, \mathbf{v}$.
- $\mathbf{u} \times \mathbf{v} = \mathbf{0} \iff \mathbf{u}, \mathbf{v}$ are parallel.
- **Anti-commutative:** $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$.

- The cross product with a fixed vector $\mathbf{u}$, i.e. $T(\mathbf{v}) = \mathbf{u} \times \mathbf{v}$, is itself a **linear transformation** $\mathbb{R}^3 \rightarrow \mathbb{R}^3$.
- This transformation can be represented by a matrix (via the "duality"/determinant trick):

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

- The **scalar triple product** $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ equals the determinant of the matrix with rows $\mathbf{a}, \mathbf{b}, \mathbf{c}$, and gives the **volume** of the parallelepiped they span.

Problem

$$\mathbf{u} = (1, 0, 0), \quad \mathbf{v} = (0, 1, 0)$$

Solution

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \hat{k} = (0, 0, 1)$$

- Magnitude = $\|\mathbf{u}\| \|\mathbf{v}\| \sin(90^\circ) = 1 = \text{area of the unit square spanned by } \hat{i}, \hat{j}.$
- Direction $(0, 0, 1)$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, confirmed by the right-hand rule.

Cramer's Rule

Statement

For the system $A\mathbf{x} = \mathbf{b}$ with $\det(A) \neq 0$, each variable x_i is given by

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is A with column i replaced by $\mathbf{b}$.

Example

$$\begin{cases} 2x + y = 5 \\ x - y = 1 \end{cases} \Rightarrow A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \det(A) = -3$$

$$x = \frac{\begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix}}{-3} = \frac{-6}{-3} = 2, \quad y = \frac{\begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix}}{-3} = \frac{-3}{-3} = 1$$

- A vector's **coordinates** depend on the chosen basis; the vector itself doesn't change.
- If $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots\}$ is a new basis, then the **change-of-basis matrix** has $\mathbf{b}_i$ as its columns.

Translating Coordinates

If $[\mathbf{v}]_B$ are the coordinates of $\mathbf{v}$ in basis B , and P is the change-of-basis matrix (columns = basis vectors expressed in standard coordinates):

$$\mathbf{v}_{\text{standard}} = P [\mathbf{v}]_B \quad \Longleftrightarrow \quad [\mathbf{v}]_B = P^{-1} \mathbf{v}_{\text{standard}}$$

- Also used to re-express a linear transformation's matrix in a new basis: $A' = P^{-1}AP$.

Problem

Express $\mathbf{v} = (3, 1)$ (standard coordinates) in the basis $B = \{\mathbf{b}_1 = (1, 1), \mathbf{b}_2 = (1, -1)\}$.

Solution

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad P^{-1} = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

$$[\mathbf{v}]_B = P^{-1}\mathbf{v} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- Check: $2\mathbf{b}_1 + 1\mathbf{b}_2 = 2(1, 1) + 1(1, -1) = (3, 1) = \mathbf{v}$. ✓

Definition

For a square matrix A , a nonzero vector $\mathbf{v}$ is an **eigenvector** with **eigenvalue** λ if

$$A\mathbf{v} = \lambda\mathbf{v}$$

- Geometrically: A only **stretches/shrinks** $\mathbf{v}$ (by factor λ) — it does **not** change its direction (except possibly a flip if $\lambda < 0$).
- Eigenvectors reveal the "natural axes" of a transformation.

Characteristic Equation

$$\det(A - \lambda I) = 0$$

Solve this polynomial equation in λ to find the eigenvalues, then solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvector.

Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

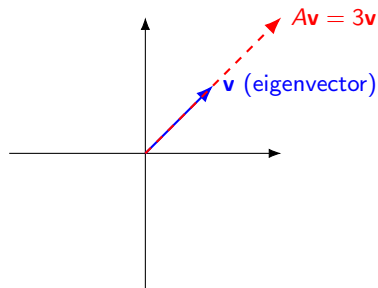
$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0 \Rightarrow \lambda = 1, 3$$

$$\text{For } \lambda = 3: (A - 3I)\mathbf{v} = 0 \Rightarrow \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda = 1: \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

- Diagonalisation: $A = PDP^{-1}$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and columns of P are the eigenvectors.

Geometric Meaning of Eigenvectors



Interpretation

- A stretches $\mathbf{v}$ by a factor of $\lambda = 3$ but keeps it on the **same line** through the origin.
- Non-eigenvectors, in general, get **rotated** as well as scaled by A .
- Eigenvalues/eigenvectors are central to PCA, stability analysis, and many ML algorithms.

General Hyperplane Equation

$$ax + by + c = 0 \text{ (line),} \quad ax + by + cz + d = 0 \text{ (plane)}$$

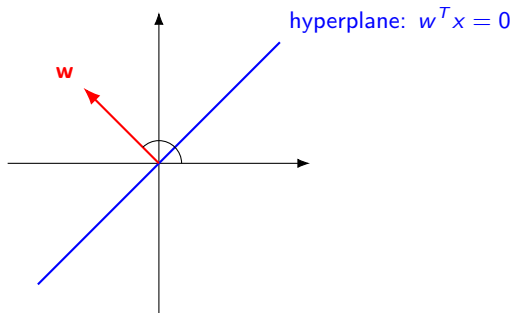
$$\sum_{i=1}^n w_i x_i + \theta = 0 \quad \Longleftrightarrow \quad \mathbf{w}^T \mathbf{x} + \theta = 0$$

Key Fact

$\mathbf{w}$ is called the **weight vector** of the hyperplane. For any hyperplane $\mathbf{w}^T \mathbf{x} = 0$, the weight vector $\mathbf{w}$ is **always perpendicular** to the hyperplane.

- Changing the weight vector $\mathbf{w}$ changes the **orientation** of the hyperplane.

Hyperplane Diagram



Interpretation

- If, for a data point $\mathbf{x}$: $\mathbf{w}^T \mathbf{x} > 0$, it lies on one side; $\mathbf{w}^T \mathbf{x} < 0$, the other side.
- This idea underlies linear classifiers (e.g. the Perceptron).

Definition

A **metric space** is a set together with a **distance function** $d(\mathbf{x}, \mathbf{y})$ that defines the distance between any two objects (points) in the set.

- Must satisfy: non-negativity, symmetry, and the triangle inequality.
- Example distance: Euclidean distance $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$.

Definition

An **inner product space** is a vector space V equipped with an inner product $\langle \mathbf{u}, \mathbf{v} \rangle$ that aligns two mathematical objects with each other.

Inner Product in $\mathbb{R}^n$

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \sum_{i=1}^n u_i v_i$$

- **Norm** of a vector: $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ — the inner product of a vector with itself.
- Inner product of **zero vectors** is 0, and they are considered **orthogonal** ($\perp$): $\langle \mathbf{u}, \mathbf{v} \rangle = 0 \Rightarrow$ orthogonal.

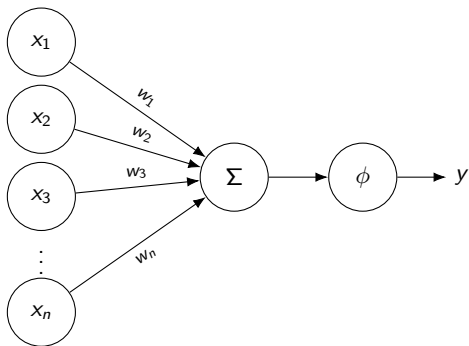
Formula

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} \implies \theta = \cos^{-1} \left[\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} \right]$$

Cosine Distance / Similarity

- Two vectors highly **similar**: $\cos \theta \rightarrow 1$ (pointing in nearly the same direction).
- Two vectors highly **dissimilar**: $\cos \theta \rightarrow 0$ (near orthogonal).
- Widely used in data science/ML (e.g. text and embedding similarity) to compare vectors regardless of magnitude.

Perceptron — The Simplest Neural Network Model



Weighted Sum and Activation

$$y = \phi \left(\sum_{i=1}^n w_i x_i - \theta \right) = \phi(\mathbf{w}^T \mathbf{x})$$

- $\mathbf{w}^T \mathbf{x}$ is exactly the **hyperplane** expression seen earlier — the perceptron classifies points based on which side of a hyperplane they fall on.
- ϕ is the **activation function** (e.g. step function outputting 0 or 1).

Output (binary classification)

$$y(\mathbf{x}) = \begin{cases} 0 \\ 1 \end{cases}$$

- This directly connects **linear algebra** (dot products, hyperplanes, weight vectors) to **machine learning** models.

Interquartile Range

$$IQR = Q_3 - Q_1$$

Outlier

Data which **stands away** from the rest of the group.

Outlier Rule (using IQR)

A data point x is an outlier if:

$$x < Q_1 - 1.5 IQR \quad \text{or} \quad x > Q_3 + 1.5 IQR$$

- Related shape measures: **Skewness** (asymmetry of the distribution) and **Kurtosis** (peakedness / tail-heaviness).

Thank You!